Pharmacology, Biochemistry and Behavior 245 (2024) 173881

Contents lists available at ScienceDirect

Pharmacology, Biochemistry and Behavior

journal homepage: www.elsevier.com/locate/pharmbiochembeh

Interaction of age and sex as factors in understanding the anxiolytic effects of alcohol: Unasked questions limiting the understanding of a critical health issue☆

Douglas B. Matthews*, Emily Kerr

Department of Psychology, University of Wisconsin – Eau Claire, Eau Claire, WI 54701, United States

ARTICLE INFO

Keywords:
Alcohol
Anxiolytic
Plus maze
Rat
Aging
Adolescent
Acute

ABSTRACT

Understanding the reasons why people consume alcohol is a critical health issue. Alcohol produces a variety of effects, including a reduction in stress or negative emotional states termed an anxiolytic effect. The anxiolytic effect of alcohol is an often-reported reason for why people begin consuming the drug. However, several factors concerning the stress-reducing effect of alcohol need to be investigated. For example, research has demonstrated that both age and sex are factors that impact alcohol's anxiolytic effect producing differential outcomes in aged female rats compared to aged male rats. In light of these findings, the current commentary highlights critical questions in need of research with the goal of better understanding how age and sex interact to influence the anxiolytic effect of alcohol. For example, the central nucleus of the amygdala has been identified as a critical brain region mediating the anxiolytic effect of drugs, but additional research is needed to understand how aging alters the neurological functioning of the central nucleus of the amygdala in both females and males. Furthermore, specific receptor isoforms, such as $GABA_A$ receptor α2, have been shown to be critical for anxiolysis and understanding how aging and sex alter receptor isoform expression by brain region is needed. Finally, age and sex interact to alter allopregnanolone levels in brain and differential neurosteroid levels may mediate alcohol's unique anxiolytic effect in aged female rats compared to aged male rats. Given the increasing age of the population in most countries and the increasing alcohol consumption levels in females compared to males, investigating the interaction of sex and age on alcohol's anxiolytic effect has great promise to discover critical answers to what are currently unasked questions.

Alcohol is one of the most used and misused drugs in the world (Ferreira and Willoughby, 2008) and alcohol consumption has occurred across most cultures spanning human history (Mandelbaum, 1965). When queried whether in research reports or in casual conversation individuals report a variety of reasons why they consume alcohol often to excess. However one of the most frequently stated reasons for alcohol drinking is to reduce stress or a negative emotional state (Beck and Treiman, 1996; Cooper et al., 2000; Keyes et al., 2012). In fact as it relates to investigating factors influencing alcohol consumption the stress-reduction hypothesis that is the anxiolytic effect of the drug has been a motivating factor which has been long proposed (Horton, 1943) and was formalized in a "tension reduction hypothesis" where alcohol consumption occurs to relieve an anxiety state (Conger, 1956). Given the negative impact of alcohol consumption on individual health (CDC fact sheet; World Health Organization 2024) and the negative financial impact on society (Sacks et al., 2015) investigators have been actively researching the biological factors that mediate the deleterious effects of alcohol on health including its anxiolytic effects. One overarching goal of this work is to reduce the misuse of alcohol as a coping strategy for negative emotional states.

When investigating the biological factors that mediate the anxiolytic effects of alcohol, animal models are often used due to ethical issues and the desire for experimental control. In most laboratories, including ours, rats and mice are used to determine factors that influence alcohol's reduction in stress. In these experiments, researchers frequently use one or more standard apparatuses which strategically pit animals' natural desire to explore novel environments against animals' natural tendency to avoid threatening locations such as brightly lit, open spaces. Typically, researchers use the elevated plus maze (Walf and Frye, 2007), open field task (Hall, 1934; Seibenhener and Wooten, 2015), or the

☆ This work was funded in part by a grant from the Mayo – UWEC Research Innovation Council.
* Corresponding author at: Department of Psychology, University of Wisconsin, Eau Claire, United States.

https://doi.org/10.1016/j.pbb.2024.173881
Received 25 June 2024; Received in revised form 6 September 2024; Accepted 11 September 2024
Available online 14 September 2024
0091-3057/

light-dark box (Crawley and Goodwin, 1980; Crawley, 1981) to investigate anxiety-like states or the anxiolytic effects of pharmacological agents. As is often found, moderate doses of alcohol produce a robust anxiolytic effect in the plus maze as measured by increased entries into the open arms or increased time in the open arms (LaBuda and Hale, 2000; LaBuda and Fuchs, 2001), food consumption in the center of the open field (Britton and Britton, 1981), and decreases in latency to reenter the light component of the light-dark box (Sharko et al., 2016). In addition to being able to document and describe alcohol's anxiolytic effect, nonhuman animal models also allow for the investigation and identification of brain regions and brain systems that produce anxiety-like behaviors in response to stress (for review see McEwen et al., 2015) along with the identification of brain systems that modulate alcohol's anxiolytic effects. For example, microinfusion of alcohol into the lateral central nucleus of the amygdala in rats produces an anxiolytic effect which is increased by nociceptin but not by CRF antagonists (Knight et al., 2020). Studies such as these highlight brain systems for investigation into alcohol's anxiolytic effects.

In the last two decades, there has been a growing awareness of the importance of both age and sex in understanding the effects of alcohol. For example, there has been much research investigating the differential effects of alcohol between adolescents and adult animals (for a review of this work see Crews et al., 2019) and a growing body of literature demonstrating unique effects of alcohol between females and males (Robinson et al., 2021). In addition, this important research has also addressed the anxiolytic effects of alcohol in animal models. For instance, it has been shown that low doses of alcohol (0.5 to 0.75 g/kg) can increase social interaction and play fighting (i.e., an anxiolytic effect) in adolescent rats but not in adult rats, suggesting developmental stage can impact the anxiolytic effects of alcohol (Sharko et al., 2016). However, unlike the differential effect of alcohol between adolescents and adult animals, similar anxiolytic effects of alcohol are found between adult male and female rats on the elevated plus maze (Wilson et al., 2004). Surprisingly though, what has not received sufficient investigation in the alcohol research field is how alcohol's effects can vary at the end of the age spectrum, namely in aged animals, and how sex can interact with aging to alter alcohol's effects.

The aged population of most counties in the world is increasing (United Nations, 2015), and older people continue to consume alcohol, often in a dangerous binge fashion (Keyes, 2023; White et al., 2023), highlighting the importance of studying alcohol's effects in the aged demographic (Matthews and Koob, 2023). In addition, the percent increase in alcohol consumption among older females is greater than males emphasizing the need to investigate sex as a factor influencing alcohol's effect in the aged demographic (McKetta and Keyes, 2019). Animal studies have consistently demonstrated that aged animals are more sensitive to the effects of alcohol than younger animals including increased cognitive, hypothermic, and ataxic impairments (Matthews and Rossmann, 2023). These results raise the possibility that aged animals may respond differentially to the anxiolytic effects of alcohol compared to younger animals. Recently, we directly tested this hypothesis by investigating the anxiolytic effect of a low dose of alcohol, 1.0 g/kg, on behavior in the elevated plus-maze of aged, adult, and adolescent female and male rats. Interestingly, we found that when normalized to control (saline exposed) animals of each age and sex, acute alcohol administration produced a significantly greater anxiolytic effect in aged female animals compared to the aged male animals or younger female animals (Matthews et al., 2024). Such a constellation of effects was surprising based on previous work. For example, we anticipated the age-dependent anxiolytic effect of acute alcohol based on previous research (Sharko et al., 2016), however, the sex-dependent effect was surprising as was the overall magnitude of the difference between aged female and aged male rats (i.e., a significant ~700 % increase in open arm time in aged females but no significant increase in aged males).

The previous work on investigating the anxiolytic effect of alcohol coupled with the recent results demonstrating that aged female rats have a significantly greater anxiolytic effect of acute alcohol compared to aged male rats raises several important questions and experimental concerns. First, there is surprisingly less research focused on investigating alcohol's anxiolytic effect in non-alcohol dependent animals compared to the anxiogenic effect found during withdrawal from chronic alcohol exposure. Given that alcohol misuse is a chronic relapsing disease, understanding factors such as anxiety during withdrawal are important as it relates to continued alcohol consumption throughout the lifespan. However, it is also important to investigate reasons why people begin drinking in the first place. As previously discussed in the current commentary, reductions in stress and negative emotional states, particularly in females, is an often-cited reason for initiating alcohol consumption (Conger, 1956; Peltier et al., 2019; Lau-Barraco and Dunn, 2009). As such, the anxiolytic effect of alcohol is important; namely, if people do not first begin consuming alcohol at high levels early in life, continued alcohol misuse resulting from withdrawal later in life would be a significantly smaller social concern. However, to further investigate this topic, it is important to theoretically conceptualize this factor correctly. As previously proposed, not only should researchers consider the anxiolytic effects of alcohol consumption as a primary motivator but must also consider the anxiolytic effects as a rewarding event, namely via negative reinforcement (Knight et al., 2020). Additional studies are needed to better understand how age, sex, life-stressors and polydrug use can impact the anxiolytic effect of alcohol directly and also as a negative reinforcer motivating continued consumption.

Secondly, it has become clear that sex and age matter. Therefore, understanding how alcohol produces differential anxiolytic effects in females compared to males and the aged population compared to the younger population are critical, understudied research fields. Our recent study (Matthews et al., 2024) began to address these issues by including age groups that spanned most of the lifespan (i.e., adolescent, young adult and aged) and both females and males. However, this study left many critical factors unanswered or not investigated. For example, it will be important for new research to validate results across multiple nonhuman animal models to ensure strain and/or species differences do not confound results. In addition, our project only used a single dose of alcohol (1.0 g/kg) which significantly limits understanding of the anxiolytic effect of the drug. Future research should use a full dose-response curve. Finally, few individuals begin drinking alcohol as an older adult and instead consume alcohol throughout their life. Longitudinal research in preclinical animal models is needed to model the complexities of drinking patterns that humans engage in to better understand the anxiolytic effects of alcohol across the lifespan in female and male subjects.

Understanding the biological mechanisms that produce significantly different alcohol-induced anxiolytic effects as a function of age and sex is an important third issue to be investigated. For example, females and males often have different blood alcohol levels even if the same amount of alcohol is consumed (human studies [Mumenthaler et al., 1999]) or administered (animal studies [Crippens et al., 1999]). However, for alcohol doses <2.5 g/kg and behavioral measurements taken within 1 h of alcohol administration in animals, blood alcohol levels are often similar between aged animals and younger animals and aged females and aged male rodents (Van Skike et al., 2010; Novier et al., 2013; Ornelas et al., 2015; Gano et al., 2017; Perkins et al., 2018; Matthews et al., 2020; Watson et al., 2020; Matthews et al., 2024). These data suggest that simple liver effects cannot explain the differences in alcohol's anxiolytic effect between aged female and aged males, further emphasizing the need for studies investigating neurobiological mechanisms that produce different responses in aged animals.

The central nucleus of the amygdala is important for stress related behaviors and mediates expression of anxiety-like behaviors in rodents (Koob and Volkow, 2016). Interestingly, the anxiolytic effect of drugs are often influenced via $GABA_A$ receptors in the central nucleus of the amygdala as demonstrated by significant anxiolytic-like effect following infusion of muscimol in rats (Sanders and Shekhar, 1995). Furthermore, low dose acute alcohol administration produces c-fos immunoreactivity in the central amygdala demonstrating this brain region is sensitive to alcohol administration (Morales et al., 1998), and as previously stated, infusion of alcohol directly into the central nucleus of the amygdala produces anxiolytic-like effects in rodents (Knight et al., 2020). Aging results in alterations of the body, including the central nervous system (Foster, 2023), in ways that can alter complex behaviors such as anxiety or anxiolytic responses to drugs such as alcohol. These results coupled with the increased anxiolytic effect in aged females compared to aged males suggest aging-related changes in central nervous system function that are sex- and age-dependent. Research into functional changes in the central nucleus of the amygdala by age and sex are needed to better understand how alcohol can produce differential anxiolytic effects in aged females and aged male animals.

It has been known for several decades that $GABA_A$ receptor structure (e.g., Hornung and Fritschy, 1996) varies by age suggesting altered pharmacology over the lifespan. For example, the distribution of $GABA_A$ receptor isoforms differs across age (Laurie et al., 1992; Fritschy and Mohler, 1995; Tobinaga et al., 2019) and brain region (Wisden et al., 1992). The differential expression pattern of $GABA_A$ receptor isoforms extend to the central amygdala where $GABA_A$ receptor α2 isoform expression appears to be highly expressed in both rodents and rhesus monkeys (Wisden et al., 1992; Marowsky et al., 2004; Sperk et al., 2020). In addition, knockin point mutations that render $GABA_A$ receptor α2 isoforms nonfunctional have shown reduced anxiolytic effects of benzodiazepine drugs suggesting this protein is critical for the anxiolytic effect of drugs (Löw et al., 2000). Given the increased anxiolytic effect of alcohol in aged female animals compared to aged male animals, future research should investigate differential expression of $GABA_A$ receptor isoform expression, particularly $GABA_A$ receptor α2 isoform expression, as a function of age and sex.

Acute alcohol exposure increases brain levels of the endogenous neurosteriod allopregnanolone, a potent $GABA_A$ receptor modulator (VanDoren et al., 2000; Torres and Ortega, 2003, 2004). As expected, acute administration of allopregnanolone produces anxiolytic effects (Crawley et al., 1986; Bitran et al., 1991) so it is likely that one mechanism by which acute alcohol exposure produces anxiolytic effects are by increasing allopregnanolone levels. Importantly, there are indications that allopregnanolone levels may also vary by age and sex. For example, allopregnanolone levels decrease as a function of age in specific rat brain regions such as the amygdala (Hasirci et al., 2017), which may produce the increased basal anxiety-like effect often seen in older rats (Matthews et al., 2024). Furthermore, serum allopregnanolone levels decrease in human males by age but not females by age (Genazzani et al., 1998). In addition, brain peripheral benzodiazepine receptors, which can synthesize allopregnanolone, increase expression in a sex-dependent manner during adolescence and young adulthood in rats and may result in differential allopregnanolone levels in adult females (Garcia and Woolley, 2008). The increase in brain peripheral benzodiazepine receptors may be a potential mechanism for increased allopregnanolone levels in the amygdala of aged animals resulting in a sex-dependent anxiolytic effect of alcohol. Based on our recent sex-dependent anxiolytic results in aged female rats (Matthews et al., 2024), research needs to determine if aged female rats have greater increases in brain allopregnanolone levels than aged male rats following acute alcohol administration, particularly in relevant brain regions such as the central nucleus of the amygdala.

Male subjects behave differently than female subjects and aged rats respond to environmental challenges in ways that are unique compared to adolescent rats. Different baseline (before an alcohol challenge) behavior(s) can produce interpretation challenges when investigating the anxiolytic effects of alcohol. In our recent project (Matthews et al., 2024), differential baseline anxiety-like scores by sex and age complicate interpretation of data. As the field begins to investigate alcohol's effects in aged animals more actively, additional tasks will likely need to be developed to better investigate the interaction of sex and age to avoid subtle confounding variables that are related to tasks and/or patterns of behavior. While many challenges can be identified to exploring the anxiolytic effect of alcohol across ages and between females and males, work has now begun to explore this critical area.

Investigation of alcohol's impact on behavior has been extensive. However, additional work investigating alcohol's anxiolytic effect is needed. Furthermore, due to the “greying” of world's population coupled with alcohol use in the demographic highlights the need for additional research investigating how age and sex interact to impact alcohol's anxiolytic effect. While timely and costly, such studies are needed to address these critical public health concerns.

CRediT authorship contribution statement

Douglas B. Matthews: Writing – original draft. **Emily Kerr:** Writing – review & editing.

Data availability

No data was used for the research described in the article.

References

Beck, K.H., Treiman, K.A., 1996. The relationship of social context of drinking, perceived social norms, and parental influence to various drinking patterns of adolescents. Addict. Behav. 21 (5), 633–644. https://doi.org/10.1016/0306-4603(95)00087-9.

Bitran, D., Hilvers, R.J., Kellogg, C.K., 1991 Oct 4. Anxiolytic effects of 3 alpha-hydroxy-5 alpha[beta]-pregnan-20-one: endogenous metabolites of progesterone that are active at the GABAA receptor. Brain Res. 561 (1), 157–161. https://doi.org/10.1016/0006-8993(91)90761-j (PMID: 1686744).

Britton, D.R., Britton, K.T., 1981 Oct. A sensitive open field measure of anxiolytic drug activity. Pharmacol. Biochem. Behav. 15 (4), 577–582. https://doi.org/10.1016/0091-3057(81)90212-4 (PMID: 6117083).

Conger, J.J., 1956. Alcoholism: theory, problem, and challenge: II. Reinforcement theory and the dynamics of alcoholism. Q. J. Stud. Alcohol 13, 296–305.

Cooper, M.L., Agocha, V.B., Sheldon, M.S., 2000 Dec. A motivational perspective on risky behaviors: the role of personality and affect regulatory processes. J. Pers. 68 (6), 1059–1088. https://doi.org/10.1111/1467-6494.00126 (PMID: 11130732).

Crawley, J.N., 1981 Nov. Neuropharmacologic specificity of a simple animal model for the behavioral actions of benzodiazepines. Pharmacol. Biochem. Behav. 15 (5), 695–699. https://doi.org/10.1016/0091-3057(81)90007-1 (PMID: 6118883).

Crawley, J., Goodwin, F.K., 1980 Aug. Preliminary report of a simple animal behavior model for the anxiolytic effects of benzodiazepines. Pharmacol. Biochem. Behav. 13 (2), 167–170. https://doi.org/10.1016/0091-3057(80)90067-2 (PMID: 6106204).

Crawley, J.N., Glowa, J.R., Majewska, M.D., Paul, S.M., 1986 Nov 29. Anxiolytic activity of an endogenous adrenal steroid. Brain Res. 398 (2), 382–385. https://doi.org/10.1016/0006-8993(86)91500-3. PMID: 2879610.

Crews, F.T., Robinson, D.L., Chandler, L.J., Ehlers, C.L., Mulholland, P.J., Pandey, S.C., Rodd, Z.A., Spear, L.P., Swartzwelder, H.S., Vetreno, R.P., 2019 Sep. Mechanisms of persistent neurobiological changes following adolescent alcohol exposure: NADIA consortium findings. Alcohol Clin. Exp. Res. 43 (9), 1806–1822. https://doi.org/10.1111/acer.14154. Epub 2019 Aug 14. PMID: 31335972; PMCID: PMC6758927.

Crippens, D., White, M.L., George, M.A., Jaworski, J.N., Brunner, L.J., Lancaster, F.E., Gonzales, R.A., 1999 Mar. Gender differences in blood levels, but not brain levels, of ethanol in rats. Alcohol. Clin. Exp. Res. 23 (3), 414–420 (PMID: 10195812).

Ferreira, M.P., Willoughby, D., 2008, Feb. Alcohol consumption: the good, the bad, and the indifferent. Appl. Physiol. Nutr. Metab. 33 (1), 12–20. https://doi.org/10.1139/H07-175.

Foster, T.C., 2023 Mar. Animal models for studies of alcohol effects on the trajectory of age-related cognitive decline. Alcohol 107, 4–11. https://doi.org/10.1016/j.alcohol.2022.04.005. Epub 2022 Apr 30. PMID: 35504438.

Fritschy, J.M., Mohler, H., 1995 Aug 14. GABAA-receptor heterogeneity in the adult rat brain: differential regional and cellular distribution of seven major subunits. J. Comp. Neurol. 359 (1), 154–194. https://doi.org/10.1002/cne.903590111 (PMID: 8557845).

Gano, A., Doremus-Fitzwater, T.L., Deak, T., 2017. A cross-sectional comparison of ethanol-related cytokine expression in the hippocampus of young and aged Fischer 344 rats. Neurobiol. Aging 54, 40–53. https://doi.org/10.1016/j.neurobiolaging.2017.01.025.

Garcia, E.F., Woolley, D.E., 2008. Gender differences in brain peripheral benzodiazepine receptor (PBR) expression and seizures produced by heptachlor during development. Proc. West. Pharmacol. Soc. 51, 18–22 (PMID: 19544667).

Genazzani, A.R., Petraglia, F., Bernardi, F., Casarosa, E., Salvestroni, C., Tonetti, A., Nappi, R.E., Luisi, S., Palumbo, M., Purdy, R.H., Luisi, M., 1998 Jun. Circulating levels of allopregnanolone in humans: gender, age, and endocrine influences. J. Clin.

Endocrinol. Metab. 83 (6), 2099–2103. https://doi.org/10.1210/jcem.83.6.4905 (PMID: 9626145).
Hall, C.S., 1934. Emotional behavior in the rat. I. Defecation and urination as measures of individual differences in emotionality. J. Comp. Psychol. 18, 385–403.
Hasirci, A.S., Maldonado-Devincci, A.M., Beattie, M.C., O'Buckley, T.K., Morrow, A.L., 2017 Feb. Cellular GABAergic neuroactive steroid (3α,5α)-3-hydroxy-pregnan-20-one (3α,5α-THP) immunostaining levels are increased in the ventral tegmental area of human alcohol use disorder patients: a postmortem study. Alcohol Clin. Exp. Res. 41 (2), 299–311. https://doi.org/10.1111/acer.13300. Epub 2017 Jan 9. PMID: 28068457; PMCID: PMC5272786.
Hornung, J.P., Fritschy, J.M., 1996 Apr 8. Developmental profile of GABAA-receptors in the marmoset monkey: expression of distinct subtypes in pre- and postnatal brain. J. Comp. Neurol. 367 (3), 413–430. https://doi.org/10.1002/(SICI)1096-9861(19960408)367:3<413::AID-CNE7>3.0.CO;2-8 (PMID: 8698901).
Horton, D.J., 1943. The function of alcohol in primitive societies. Q. J. Stud. Alcohol 4, 199–320.
Keyes, K.M., 2023. Alcohol use in the older adult US population: trends, causes, and consequences. Alcohol 107, 28–31. https://doi.org/10.1016/j.alcohol.2022.05.005.
Keyes, K.M., Hatzenbuehler, M.L., Grant, B.F., Hasin, D.S., 2012. Stress and alcohol: epidemiologic evidence. Alcohol Res. 34 (4), 391–400 (PMID: 23584105; PMCID: PMC3797525).
Knight, C.P., Hauser, S.R., Waeiss, R.A., Molosh, A.I., Johnson, P.L., Truitt, W.A., McBride, W.J., Bell, R.L., Shekhar, A., Rodd, Z.A., 2020 Sep. The rewarding and anxiolytic properties of ethanol within the central nucleus of the amygdala: mediated by genetic background and nociceptin. J. Pharmacol. Exp. Ther. 374 (3), 366–375. https://doi.org/10.1124/jpet.119.262097 (Epub 2020 Jun 11. PMID: 32527792; PMCID: PMC7430446).
Koob, G.F., Volkow, N.D., 2016 Aug. Neurobiology of addiction: a neurocircuitry analysis. Lancet Psychiatry 3 (8), 760–773. https://doi.org/10.1016/S2215-0366(16)00104-8. PMID: 27475769; PMCID: PMC6135092.
LaBuda, C.J., Fuchs, P.N., 2001 Aug. The anxiolytic effect of acute ethanol or diazepam exposure is unaltered in mu-opioid receptor knockout mice. Brain Res. Bull. 55 (6), 755–760. https://doi.org/10.1016/s0361-9230(01)00569-x (PMID: 11595359).
LaBuda, C.J., Hale, R.L., 2000 Jan. Anxiety in mice following acute aspartame and ethanol exposure. Alcohol 20 (1), 69–74. https://doi.org/10.1016/s0741-8329(99)00060-9 (PMID: 10680719).
Lau-Barraco, C., Dunn, M.E., 2009. Environmental context effects on alcohol cognitions and immediate alcohol consumption. Addict. Res. Theory 17 (3), 306–314. https://doi.org/10.1080/16066350802346201.
Laurie, D.J., Wisden, W., Seeburg, P.H., 1992 Nov. The distribution of thirteen GABAA receptor subunit mRNAs in the rat brain. III. Embryonic and postnatal development. J. Neurosci. 12 (11), 4151–4172. https://doi.org/10.1523/JNEUROSCI.12-11-04151.1992 (PMID: 1331359; PMCID: PMC6576006).
Löw, K., Crestani, F., Keist, R., Benke, D., Brünig, I., Benson, J.A., Fritschy, J.M., Rülicke, T., Bluethmann, H., Möhler, H., Rudolph, U., 2000 Oct 6. Molecular and neuronal substrate for the selective attenuation of anxiety. Science 290 (5489), 131–134. https://doi.org/10.1126/science.290.5489.131. Erratum in: Science 2000 Nov 3;290(5493):936c. PMID: 11021797.
Mandelbaum, D.G., 1965. Alcohol and Culture. Cur. Anthro. 6 (3), 281–293.
Marowsky, A., Fritschy, J.M., Vogt, K.E., 2004 Sep. Functional mapping of GABA A receptor subtypes in the amygdala. Eur. J. Neurosci. 20 (5), 1281–1289. https://doi.org/10.1111/j.1460-9568.2004.03574.x (PMID: 15341600).
Matthews, D.B., Koob, G.F., 2023. On the critical need to investigate the effect of alcohol in the older population. Alcohol 107, 2–3. https://doi.org/10.1016/j.alcohol.2022.09.002.
Matthews, D.B., Rossmann, G., 2023. Using animal models to identify clinical risk factors in the older population due to alcohol use and misuse. Alcohol 107, 38–43. https://doi.org/10.1016/j.alcohol.2022.05.003.
Matthews, D.B., Scaletty, S., Schreiber, A., Trapp, S., 2020. Acute ethanol administration produces larger spatial and nonspatial memory impairments in 29–33 month old rats compared to adult and 18–24 month old rats. Pharmacol. Biochem. Behav. 199, 173074 https://doi.org/10.1016/j.pbb.2020.173074.
Matthews, D.B., Rossmann, G., Matthews, S.J., Zank, A., Shult, C., Turunen, A., Sharma, P., 2024 Jun. Increased anxiolytic effect in aged female rats and increased motoric behavior in aged male rats to acute alcohol administration: comparison to younger animals. Pharmacol. Biochem. Behav. 239, 173770 https://doi.org/10.1016/j.pbb.2024.173770. Epub 2024 Apr 16. PMID: 38636813.
McEwen, B.S., Bowles, N.P., Gray, J.D., Hill, M.N., Hunter, R.G., Karatsoreos, I.N., Nasca, C., 2015 Oct. Mechanisms of stress in the brain. Nat. Neurosci. 18 (10), 1353–1363. https://doi.org/10.1038/nn.4086. Epub 2015 Sep 25. PMID: 26404710; PMCID: PMC4933289.
McKetta, S., Keyes, K.M., 2019. Heavy and binge alcohol drinking and parenting status in the United States from 2006 to 2018: an analysis of nationally representative cross-sectional surveys. PLoS Med. 16 (11), e1002954 https://doi.org/10.1371/journal.pmed.1002954.
Morales, M., Criado, J.R., Sanna, P.P., Henriksen, S.J., Bloom, F.E., 1998 Jul 6. Acute ethanol induces c-fos immunoreactivity in GABAergic neurons of the central nucleus of the amygdala. Brain Res. 798 (1–2), 333–336. https://doi.org/10.1016/s0006-8993(98)00457-0 (PMID: 9666163).
Mumenthaler, M.S., Taylor, J.L., O'Hara, R., Yesavage, J.A., 1999. Gender differences in moderate drinking effects. Alcohol Res. Health 23 (1), 55–64 (PMID: 10890798; PMCID: PMC6761697).
Novier, A., Van Skike, C.E., Diaz-Granados, J.L., Mittleman, G., Matthews, D.B., 2013. Acute alcohol produces ataxia and cognitive impairments in aged animals: a comparison between young adult and aged rats. Alcohol. Clin. Exp. Res. 37 (8), 1317–1324. https://doi.org/10.1111/acer.12110.
Ornelas, L.C., Novier, A., Van Skike, C.E., Diaz-Granados, J.L., Matthews, D.B., 2015. The effects of acute alcohol on motor impairments in adolescent, adult, and aged rats. Alcohol 49 (2), 121–126. https://doi.org/10.1016/j.alcohol.2014.12.002.
Peltier, M.R., Verplaetse, T.L., Mineur, Y.S., Petrakis, I.L., Cosgrove, K.P., Picciotto, M.R., McKee, S.A., 2019 Feb 8. Sex differences in stress-related alcohol use. Neurobiol. Stress 10, 100149. https://doi.org/10.1016/j.ynstr.2019.100149. PMID: 30949562; PMCID: PMC6430711.
Perkins, A.E., Vore, A.S., Lovelock, D., Varlinskaya, E., Deak, T., 2018. Late aging alters behavioral sensitivity to ethanol in a sex-specific manner in Fischer 344 rats. Pharmacol. Biochem. Behav. 175, 1–9. https://doi.org/10.1016/j.pbb.2018.07.012.
Robinson, D.L., Amodeo, L.R., Chandler, L.J., Crews, F.T., Ehlers, C.L., Gómez-A, A., Healey, K.L., Kuhn, C.M., Macht, V.A., Marshall, S.A., Swartzwelder, H.S., Varlinskaya, E.I., Werner, D.F., 2021. The role of sex in the persistent effects of adolescent alcohol exposure on behavior and neurobiology in rodents. Int. Rev. Neurobiol. 160, 305–340. https://doi.org/10.1016/bs.irn.2021.07.007. Epub 2021 Aug 11. PMID: 34696877; PMCID: PMC8672816.
Sacks, J.J., Gonzales, K.R., Bouchery, E.E., et al., 2015. 2010 national and state costs of excessive alcohol consumption. Am. J. Prev. Med. 49, e73–e79.
Sanders, S.K., Shekhar, A., 1995 Dec. Regulation of anxiety by GABAA receptors in the rat amygdala. Pharmacol. Biochem. Behav. 52 (4), 701–706. https://doi.org/10.1016/0091-3057(95)00153-n (PMID: 8587908).
Seibenhener, M.L., Wooten, M.C., 2015 Feb 6. Use of the open field maze to measure locomotor and anxiety-like behavior in mice. J. Vis. Exp. 96, e52434 doi: 10.3791/52434. PMID: 25742564; PMCID: PMC4354627.
Sharko, A.C., Kaigler, K.F., Fadel, J.R., Wilson, M.A., 2016 Feb. Ethanol-induced anxiolysis and neuronal activation in the amygdala and bed nucleus of the stria terminalis. Alcohol 50, 19–25. https://doi.org/10.1016/j.alcohol.2015.11.001. Epub 2015 Nov 26. PMID: 26775553; PMCID: PMC4753081.
Sperk, G., Kirchmair, E., Bakker, J., Sieghart, W., Drexel, M., Kondova, I., 2020 Oct 15. Immunohistochemical distribution of 10 $GABA_A$ receptor subunits in the forebrain of the rhesus monkey Macaca mulatta. J. Comp. Neurol. 528 (15), 2551–2568. https://doi.org/10.1002/cne.24910 (Epub 2020 Apr 3. PMID: 32220012; PMCID: PMC7496627).
Tobinaga, M., Suzuki, Y., Fujinaka, H., Ozawa, T., Nakajima, T., 2019 Jun. Age-related increase in GABAA receptor distribution in the prefrontal cortex. J. Clin. Neurosci. 64, 106–110. https://doi.org/10.1016/j.jocn.2019.03.044. Epub 2019 Apr 1. PMID: 30948309.
Torres, J.M., Ortega, E., 2003 Aug. Differential regulation of steroid 5alpha-reductase isozymes expression by androgens in the adult rat brain. FASEB J. 17 (11), 1428–1433. https://doi.org/10.1096/fj.02-1119com (PMID: 12890696).
Torres, J.M., Ortega, E., 2004 Mar. Alcohol intoxication increases allopregnanolone levels in male adolescent humans. Psychopharmacology (Berl) 172 (3), 352–355. https://doi.org/10.1007/s00213-003-1662-0. Epub 2003 Nov 28. PMID: 14647956.
United Nations, 2015. The World Population Prospects, 2015 Revision.
Van Skike, C.E., Botta, P., Chin, V.S., et al., 2010. Behavioral effects of ethanol in cerebellum are age dependent: potential system and molecular mechanisms. Alcohol. Clin. Exp. Res. 34 (12), 2070–2080. https://doi.org/10.1111/j.1530-0277.2010.01303.x.
VanDoren, M.J., Matthews, D.B., Janis, G.C., Grobin, A.C., Devaud, L.L., Morrow, A.L., 2000 Mar 1. Neuroactive steroid 3alpha-hydroxy-5alpha-pregnan-20-one modulates electrophysiological and behavioral actions of ethanol. J. Neurosci. 20 (5), 1982–1989 doi: 10.1523/JNEUROSCI.20-05-01982.2000. PMID: 10684899; PMCID: PMC6772919.
Walf, A.A., Frye, C.A., 2007. The use of the elevated plus maze as an assay of anxiety-related behavior in rodents. Nat. Protoc. 2 (2), 322–328. https://doi.org/10.1038/nprot.2007.44.
Watson, M.R., James, K., Mittleman, G., Matthews, D.B., 2020. Impact of acute ethanol exposure on body temperatures in aged, adult and adolescent male rats. Alcohol 82, 81–89. https://doi.org/10.1016/j.alcohol.2019.08.001.
White, A.M., Orosz, A., Powell, P.A., Koob, G.F., 2023. Alcohol and aging - an area of increasing concern. Alcohol 107, 19–27. https://doi.org/10.1016/j.alcohol.2022.07.005.
Wilson, M.A., Burghardt, P.R., Ford, K.A., Wilkinson, M.B., Primeaux, S.D., 2004 Jul. Anxiolytic effects of diazepam and ethanol in two behavioral models: comparison of males and females. Pharmacol. Biochem. Behav. 78 (3), 445–458. https://doi.org/10.1016/j.pbb.2004.04.017 (PMID: 15251253).
Wisden, W., Laurie, D.J., Monyer, H., Seeburg, P.H., 1992 Mar. The distribution of 13 GABAA receptor subunit mRNAs in the rat brain. I. Telencephalon, diencephalon, mesencephalon. J. Neurosci. 12 (3), 1040–1062. https://doi.org/10.1523/JNEUROSCI.12-03-01040.1992. PMID: 1312131; PMCID: PMC6576059.